

SMART WOMEN SAFETY & EMERGENCY RESPONSE APPLICATION

A

MAJOR PROJECT-II REPORT

Submitted in partial fulfillment of the requirement.

for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

By

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June - 2026

Sagar Institute of Science & Technology (SISTec), Bhopal (M.P)

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CERTIFICATE

We hereby certify that the work which is being presented in the B.Tech. Major Project-II Report entitled **SMART WOMEN SAFETY & EMERGENCY RESPONSE APPLICATION**, in partial fulfillment of the requirements for the award of the degree of *Bachelor of Technology*, submitted to the Department of **Computer Science & Engineering**, Sagar Institute of Science & Technology (SISTec), Bhopal (M.P.) is an authentic record of our own work carried out during the period from Jan-2026 to Jun-2026 under the supervision of **Dr. Komal Tahlilani**.

The content presented in this project has not been submitted by me for the award of any other degree elsewhere.

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ABSTRACT

The Smart Women Safety & Emergency Response Application is a mobile-based solution designed to enhance the safety and security of women by providing instant assistance during emergency situations. The application integrates advanced features such as GPS-based live location tracking, SOS alerts, voice activation, and real-time communication with pre-registered contacts and emergency services, ensuring quick response and support. It may also include functionalities like identifying nearby safe locations, accessing emergency helplines, and storing evidence for future use. By reducing response time and enabling proactive safety measures, the system empowers women to move freely with greater confidence and independence. This project aligns with the goals of the United Nations, particularly **SDG 5: Gender Equality** by promoting women's empowerment, **SDG 9: Industry, Innovation and Infrastructure** through the use of innovative technology, and **SDG 16: Peace, Justice and Strong Institutions** by contributing to a safer and more secure society.

LIST OF ABBREVIATIONS

<u>ACRONYM</u>	<u>FULL FORM</u>
SDLC	Software Development Life Cycle
SDK	Software Development Kit
SQL	Structured Query Language
HTML	Hypertext Markup Language
UML	Unified Modelling Language

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Chapter 1

Introduction

CHAPTER 3

PROBLEM DESCRIPTION

3.1 INTRODUCTION

Safety of women is a major concern in today's society. Despite various laws and awareness programs, incidents of harassment, assault, and unsafe situations continue to occur in both urban and rural areas. Women often face insecurity while traveling, working late hours, or even performing daily activities. This situation highlights the need for an efficient system that can provide immediate help during emergency. The **women safety application** is developed as a technological solution to address these concerns. This application is designed to provide immediate assistance to women during emergency situations. It utilizes modern technologies such as mobile communication, gps tracking, and real-time notifications to ensure quick response and support.

3.2 ANALYSIS OF CURRENT PRACTICES

Currently, women rely on traditional methods for ensuring their safety, such as making phone calls to family or friends, manually informing others about their location, depending on nearby people for help, and using basic emergency helpline numbers. However, these methods have several limitations, especially during critical situations. They are often time-consuming and require manual effort to communicate important details like location. In emergency scenarios, panic and stress can make it difficult to take quick action, and network issues may further reduce reliability. Additionally, these approaches lack real-time tracking capabilities and do not provide any automated alert mechanism, which can delay response time and reduce the chances of receiving immediate assistance.

3.3 PROBLEM IDENTIFICATION

The major problems identified in the existing system are:

- Delay in sending emergency alerts
- Difficulty in sharing exact location
- Lack of immediate response mechanism
- Dependency on manual communication

- Absence of centralized safety system

These issues can increase the risk and make it difficult for victims to get timely assistance.

3.4 PROPOSED SYSTEM

To overcome the limitations of the existing system, a **Women Safety Application** is proposed.

This system provides:

- One-click SOS alert feature
- Real-time GPS location sharing
- Automatic notification to emergency contacts
- Quick and reliable communication system

The application ensures that help can be reached faster and more efficiently.

3.5 FEATURES OF THE PROPOSED SYSTEM

- Emergency SOS button
- Live location tracking
- Instant alert message to contacts
- Easy-to-use interface
- Secure user data management
- Optional audio/video recording

3.6 ADVANTAGES OF PROPOSED SYSTEM

- Reduces response time during emergencies
- Provides accurate location tracking
- Minimizes manual effort
- Improves safety and confidence of users
- Works efficiently in critical situations

Proposed Methodology

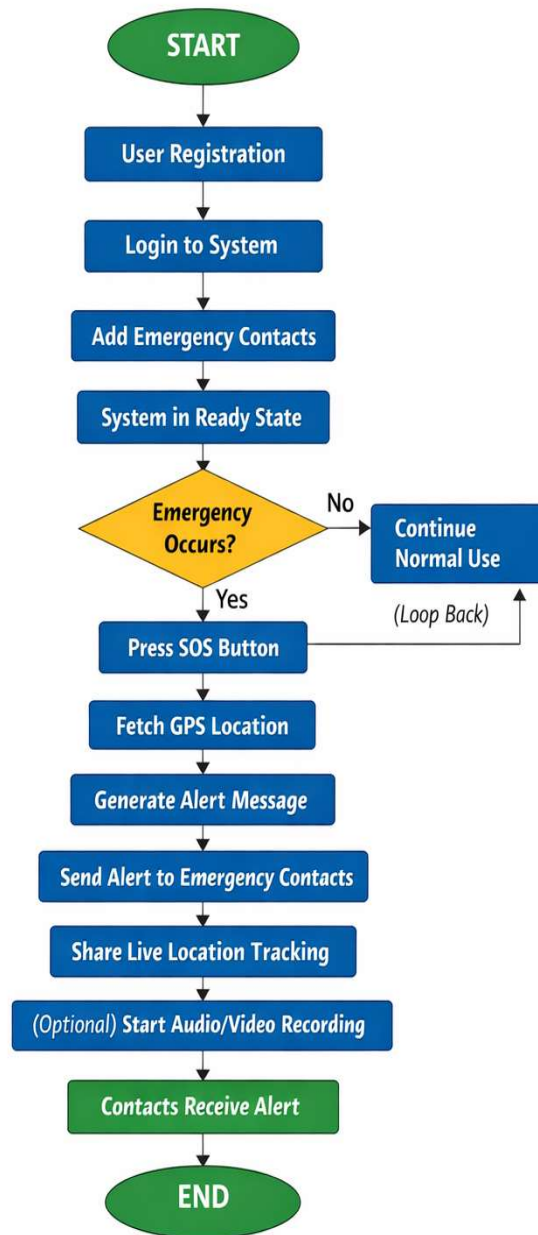


Figure 3.1: Proposed Methodology

3.7 FINAL OUTCOME

The existing methods for ensuring women's safety are not sufficient to handle emergency situations effectively. There is a clear need for a smart and automated solution. The proposed Women Safety Application addresses these challenges by providing a fast, reliable, and user-friendly system that enhances personal security and ensures timely assistance.

Table 3.1: Functional Requirement Specification

Req ID	Module	Requirement Description	Input	Output
FR1	User Registration	User should be able to register using basic details	Name, Email, Phone	Account created
FR2	User Login	User should login using credentials/OTP	Username, Password	Access system
FR3	Manage Contacts	User can add/edit/delete emergency contacts	Contact details	Updated contact list
FR4	SOS Alert	User can trigger emergency alert using SOS button	SOS trigger	Alert sent to contacts
FR5	Location Tracking	System should fetch real-time GPS location	GPS data	Location coordinates

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PROJECT SUMMARY

About Project

Title of the project	
Semester	
Members	
Team Leader	
Describe role of every member in the project	
What is the motivation for selecting this project?	
Project Type (Desktop Application, Web Application, Mobile App, Web)	

Tools & Technologies

Programming language used	
Compiler used (with version)	
IDE used (with version)	
Front End Technologies (with version, wherever Applicable)	
Back End Technologies (with version, wherever applicable)	
Database used (with version)	

Software Design& Coding

Is prototype of the software developed?	
SDLC model followed (Waterfall, Agile, Spiral etc.)	
Why above SDLC model is followed?	
Justify that the SDLC model mentioned above is followed in the project.	
Software Design approach followed (Functional or Object-oriented)	
Name the diagrams developed (According to the Design approach followed)	
In case Object Oriented approach is followed, which of the OOPS principles are covered in design?	
No. of Tiers (example 3-tier)	
Total no. of front-end pages	
Total no. of tables in database	
Database is in which Normal Form?	
Are the entries in database encrypted?	
Front end validations applied (Yes / No)	
Session management done (in case of web applications)	

Is application browser compatible (in case of web applications)	
Exception handling done (Yes / No)	
Commenting done in code (Yes / No)	
Naming convention followed (Yes / No)	
What difficulties faced during deployment of project?	
Total no. of Use-cases	
Give titles of Use-cases	

Project Requirements

MVC architecture followed (Yes / No)	
If yes, write the name of MVC architecture followed (MVC-1, MVC-2)	
Design Pattern used (Yes / No)	
If yes, write the name of Design Pattern used	
Interface type (CLI / GUI)	
No. of Actors	
Name of Actors	
Total no. of Functional Requirements	
List few important non- Functional Requirements	

Testing

Which test is performed? (Manual or Automation)	
Is Beta testing done for this project?	

Write project narrative covering above mentioned points

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Guide Signature
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APPENDIX-1

GLOSSARY OF TERMS

(In alphabetical order)

B

- BER** Bit Error Rate. The number of bits received in error. Usually, the number is quite low and expressed as a ratio in scientific notation. 10^{-2} means one-bit in 100 is received in error.
- BPSK** Binary Phase Shift Keying. A modulation method that encodes bits as phase shifts. One of two phase shifts can be selected to encode a single bit.
- BSS** Basic Service Set. The building block of 802.11 networks. A BSS is a set of stations that are logically associated with each other.
- BSSID** Basic Service Set Identifier. A 48-bit identifier used by all stations in a BSS in frame headers.

C

- CCITT** Comité Consultatif International Télégraphique et Téléphonique. A UN body responsible for telephone standardization. Due to a reorganization, it is now called the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T).
- CCK** Complementary Code Keying. A modulation scheme that transforms data blocks into complex codes and is capable of encoding several bits per block.
- CF** Contention Free. Services that do not involve contention for the medium are contention-free services. Such services are implemented by a Point Coordinator (PC) through the use of the Point Coordination Function (PCF). Contention-free services are not widely implemented.